Model Architecture: Llama-3.2-1B

Generated by sg_modelload

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1 Model Overview

Model name	Llama-3.2-1B
Transformers class	`LlamaForCausalLM`
Path	`/models/memorization/Llama-3.2-1B`
Total parameters	1,235,814,400 (1.24B)
Weight dtype	bfloat16
Hidden size d	2,048
Decoder layers L	16
Query heads H	32
Key/value heads H_{kv}	8
Head dim d_h	64
MLP width d_{ff}	8,192
Vocabulary V	128,256
Max context length	131,072
RMSNorm ϵ	1e-05
Tied input/output embeddings	True
Activation	SiLU
RoPE base θ	500,000

How to read this document. Section 2 writes the whole network as one composed function. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Notation

B, T	batch size, sequence length (tokens)
d	hidden size (2048)
V	vocabulary size (128,256)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 The Model as a Function

Input: token ids $x \in \{0, \ldots, V-1\}^{B \times T}$. Output: next-token logits $y \in \mathbb{R}^{B \times T \times V}$. The 16 decoder layers are written with composition powers, so a block that repeats n times appears once as $\mathcal{B}^{\circ n}$.

$$
\begin{aligned}
h_0 &= \text{Embed}(x) \\
h_L &= \left(\mathcal{B}_{\text{D}}^{\circ 16}\right)(h_0) \\
y &= W_E^{\top}\ \text{RMSNorm}_{\text{final}}(h_L)
\end{aligned}
$$

2.1 Layer pattern

Symbol	Block type	Count	Params / block	Layer indices
$\mathcal{B}_{\text{D}}$	Decoder block	16	60,821,504	0–15

Language-model decoder (16 layers). layers 0–15: $[\mathcal{B}_{\text{D}}]$ repeated 16×.

2.2 Block functions

$\mathcal{B}_{\mathrm{D}}$ — **Decoder block.**

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
Q &= W_Q\, u && H = 32 \text{ heads of size } d_h = 64\\
K,\ V &= W_K\, u, \quad W_V\, u && H_{kv} = 8 \text{ key/value heads}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)} &&\\
\mathrm{Attn}(u) &= W_O\,[O_1; \ldots; O_H] &&\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\mathrm{SiLU}(W_{\mathrm{gate}}\, u) \odot W_{\mathrm{up}}\, u\big) && \text{gated MLP, width } 8192
\end{aligned}
$$

3 Forward Pass in Pseudocode

Algorithm 1 Llama-3.2-1B: full forward pass

Require: token ids x of shape $B \times T$

$h \leftarrow \mathrm{Embed}(x)$ ▷ look up one vector per token: $B{\times}T{\times}2048$
for $r \leftarrow 0$ **to** 15 **do** ▷ layers 0–15: pattern of 1 repeated 16 times
  $h \leftarrow \mathcal{B}_{\mathrm{D}}(h)$ ▷ Decoder block
end for
$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$
$y \leftarrow h\, W_E^{\top}$ ▷ logits $B{\times}T{\times}128,256$, embedding matrix re-used
return y ▷ softmax(y) = next-token probabilities

Algorithm 2 Block $\mathcal{B}_{\mathrm{D}}(h)$ — Decoder block

Require: h of shape $B \times T \times 2048$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^{\top}$ ▷ $B{\times}T{\times}32{\times}64$ (32 query heads)
$K \leftarrow uW_K^{\top};\ V \leftarrow uW_V^{\top}$ ▷ $B{\times}T{\times}8{\times}64$ (8 key/value heads, each shared by 4 query heads)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\, QK^{\top} + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}32{\times}64$
$m \leftarrow O\, W_O^{\top}$ ▷ concatenate heads, project back to $B{\times}T{\times}2048$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^{\top}) \odot uW_{\mathrm{up}}^{\top}\big)W_{\mathrm{down}}^{\top}$ ▷ expand to 8192, gate, shrink back to 2048
$h \leftarrow r + m$ ▷ add residual
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 2: Weights that appear once

Weight	Type	Shape	Params
`model.embed_tokens.weight`	Embedding	128256×2048	262,668,288
`model.norm.weight`	LlamaRMSNorm	2048	2,048

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Weight	Type	Shape	Params
`lm_head.weight` (tied to `model.embed_tokens.weight`)	Linear	128256×2048	0 (shared)

4.2 Language-model decoder: `model.layers`

Table 3: $\mathcal{B}_{\mathrm{D}}$: Decoder block — layers 0–15; weight names are relative to `model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	2048×2048	4,194,304
`self_attn.k_proj.weight`	Linear	512×2048	1,048,576
`self_attn.v_proj.weight`	Linear	512×2048	1,048,576
`self_attn.o_proj.weight`	Linear	2048×2048	4,194,304
`mlp.gate_proj.weight`	Linear	8192×2048	16,777,216
`mlp.up_proj.weight`	Linear	8192×2048	16,777,216
`mlp.down_proj.weight`	Linear	2048×8192	16,777,216
`input_layernorm.weight`	LlamaRMSNorm	2048	2,048
`post_attention_layernorm.weight`	LlamaRMSNorm	2048	2,048
Per block			**60,821,504**
× 16 blocks			**973,144,064**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
Language model	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.

6 Parameter Count

Table 5: Parameters by component

Component	Parameters	% of total
MLP / MoE	805,306,368	65.2%
Token embedding	262,668,288	21.3%
Attention / token mixer	167,772,160	13.6%
Norms	67,584	0.0%
Total	**1,235,814,400**	**100.0%**

Table 6: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{D}}$ Decoder block	16	60,821,504	973,144,064
Modules outside stacks	1	–	262,670,336
Total			**1,235,814,400**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight` *(tied)*	Linear	128256 × 2048	bfloat16	262,668,288
`model`	LlamaModel	–	–	–
`model.embed_tokens.weight`	Embedding	128256 × 2048	bfloat16	262,668,288
`model.layers.0`	LlamaDecoderLayer	–	–	–
`model.layers.0.input_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.0.mlp`	LlamaMLP	–	–	–
`model.layers.0.mlp.down_proj.weight`	Linear	2048 × 8192	bfloat16	16,777,216
`model.layers.0.mlp.gate_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.0.mlp.up_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.0.post_attention_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.0.self_attn`	LlamaAttention	–	–	–
`model.layers.0.self_attn.k_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.0.self_attn.o_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.0.self_attn.q_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.0.self_attn.v_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.1`	LlamaDecoderLayer	–	–	–
`model.layers.1.input_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.1.mlp`	LlamaMLP	–	–	–
`model.layers.1.mlp.down_proj.weight`	Linear	2048 × 8192	bfloat16	16,777,216
`model.layers.1.mlp.gate_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.1.mlp.up_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.1.post_attention_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.1.self_attn`	LlamaAttention	–	–	–
`model.layers.1.self_attn.k_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.1.self_attn.o_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.1.self_attn.q_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.1.self_attn.v_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.2`	LlamaDecoderLayer	–	–	–
`model.layers.2.input_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.2.mlp`	LlamaMLP	–	–	–
`model.layers.2.mlp.down_proj.weight`	Linear	2048 × 8192	bfloat16	16,777,216
`model.layers.2.mlp.gate_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.2.mlp.up_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.2.post_attention_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.2.self_attn`	LlamaAttention	–	–	–
`model.layers.2.self_attn.k_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.2.self_attn.o_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.2.self_attn.q_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.2.self_attn.v_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.3`	LlamaDecoderLayer	–	–	–
`model.layers.3.input_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.3.mlp`	LlamaMLP	–	–	–
`model.layers.3.mlp.down_proj.weight`	Linear	2048 × 8192	bfloat16	16,777,216

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Layer path	Type	Shape	Dtype	Params
model.layers.3.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.3.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.3.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.3.self_attn	LlamaAttention	–	–	–
model.layers.3.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.3.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.3.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.3.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.4	LlamaDecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.4.mlp	LlamaMLP	–	–	–
model.layers.4.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.4.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.4.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.4.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.4.self_attn	LlamaAttention	–	–	–
model.layers.4.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.4.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.4.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.4.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.5	LlamaDecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.5.mlp	LlamaMLP	–	–	–
model.layers.5.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.5.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.5.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.5.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.5.self_attn	LlamaAttention	–	–	–
model.layers.5.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.5.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.5.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.5.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.6	LlamaDecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.6.mlp	LlamaMLP	–	–	–
model.layers.6.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.6.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.6.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.6.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.6.self_attn	LlamaAttention	–	–	–
model.layers.6.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.6.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.6.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.6.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.7	LlamaDecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.7.mlp	LlamaMLP	–	–	–
model.layers.7.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.7.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.7.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.7.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.7.self_attn	LlamaAttention	–	–	–
model.layers.7.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.7.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.7.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.7.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.8	LlamaDecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.8.mlp	LlamaMLP	–	–	–
model.layers.8.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.8.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.8.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.8.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.8.self_attn	LlamaAttention	–	–	–
model.layers.8.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.8.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.8.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.8.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576

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Layer path	Type	Shape	Dtype	Params
model.layers.9	LlamaDecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.9.mlp	LlamaMLP	–	–	–
model.layers.9.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.9.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.9.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.9.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.9.self_attn	LlamaAttention	–	–	–
model.layers.9.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.9.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.9.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.9.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.10	LlamaDecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.10.mlp	LlamaMLP	–	–	–
model.layers.10.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.10.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.10.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.10.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.10.self_attn	LlamaAttention	–	–	–
model.layers.10.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.10.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.10.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.10.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.11	LlamaDecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.11.mlp	LlamaMLP	–	–	–
model.layers.11.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.11.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.11.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.11.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.11.self_attn	LlamaAttention	–	–	–
model.layers.11.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.11.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.11.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.11.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.12	LlamaDecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.12.mlp	LlamaMLP	–	–	–
model.layers.12.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.12.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.12.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.12.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.12.self_attn	LlamaAttention	–	–	–
model.layers.12.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.12.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.12.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.12.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.13	LlamaDecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.13.mlp	LlamaMLP	–	–	–
model.layers.13.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.13.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.13.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.13.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.13.self_attn	LlamaAttention	–	–	–
model.layers.13.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.13.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.13.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.13.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.14	LlamaDecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.14.mlp	LlamaMLP	–	–	–
model.layers.14.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.14.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.14.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.14.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.14.self_attn	LlamaAttention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.layers.14.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.14.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.14.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.14.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.15	LlamaDecoderLayer	–	–	–
model.layers.15.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.15.mlp	LlamaMLP	–	–	–
model.layers.15.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.15.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.15.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.15.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.15.self_attn	LlamaAttention	–	–	–
model.layers.15.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.15.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.15.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.15.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.norm.weight	LlamaRMSNorm	2048	bfloat16	2,048

Appendix B Per-Layer Parameter Count

Module	Params	Running total
model.embed_tokens	262,668,288	262,668,288
model.layers.0.input_layernorm	2,048	262,670,336
model.layers.0.mlp.down_proj	16,777,216	279,447,552
model.layers.0.mlp.gate_proj	16,777,216	296,224,768
model.layers.0.mlp.up_proj	16,777,216	313,001,984
model.layers.0.post_attention_layernorm	2,048	313,004,032
model.layers.0.self_attn.k_proj	1,048,576	314,052,608
model.layers.0.self_attn.o_proj	4,194,304	318,246,912
model.layers.0.self_attn.q_proj	4,194,304	322,441,216
model.layers.0.self_attn.v_proj	1,048,576	323,489,792
model.layers.1.input_layernorm	2,048	323,491,840
model.layers.1.mlp.down_proj	16,777,216	340,269,056
model.layers.1.mlp.gate_proj	16,777,216	357,046,272
model.layers.1.mlp.up_proj	16,777,216	373,823,488
model.layers.1.post_attention_layernorm	2,048	373,825,536
model.layers.1.self_attn.k_proj	1,048,576	374,874,112
model.layers.1.self_attn.o_proj	4,194,304	379,068,416
model.layers.1.self_attn.q_proj	4,194,304	383,262,720
model.layers.1.self_attn.v_proj	1,048,576	384,311,296
model.layers.2.input_layernorm	2,048	384,313,344
model.layers.2.mlp.down_proj	16,777,216	401,090,560
model.layers.2.mlp.gate_proj	16,777,216	417,867,776
model.layers.2.mlp.up_proj	16,777,216	434,644,992
model.layers.2.post_attention_layernorm	2,048	434,647,040
model.layers.2.self_attn.k_proj	1,048,576	435,695,616
model.layers.2.self_attn.o_proj	4,194,304	439,889,920
model.layers.2.self_attn.q_proj	4,194,304	444,084,224
model.layers.2.self_attn.v_proj	1,048,576	445,132,800
model.layers.3.input_layernorm	2,048	445,134,848
model.layers.3.mlp.down_proj	16,777,216	461,912,064
model.layers.3.mlp.gate_proj	16,777,216	478,689,280
model.layers.3.mlp.up_proj	16,777,216	495,466,496
model.layers.3.post_attention_layernorm	2,048	495,468,544
model.layers.3.self_attn.k_proj	1,048,576	496,517,120
model.layers.3.self_attn.o_proj	4,194,304	500,711,424
model.layers.3.self_attn.q_proj	4,194,304	504,905,728
model.layers.3.self_attn.v_proj	1,048,576	505,954,304
model.layers.4.input_layernorm	2,048	505,956,352

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Module	Params	Running total
model.layers.4.mlp.down_proj	16,777,216	522,733,568
model.layers.4.mlp.gate_proj	16,777,216	539,510,784
model.layers.4.mlp.up_proj	16,777,216	556,288,000
model.layers.4.post_attention_layernorm	2,048	556,290,048
model.layers.4.self_attn.k_proj	1,048,576	557,338,624
model.layers.4.self_attn.o_proj	4,194,304	561,532,928
model.layers.4.self_attn.q_proj	4,194,304	565,727,232
model.layers.4.self_attn.v_proj	1,048,576	566,775,808
model.layers.5.input_layernorm	2,048	566,777,856
model.layers.5.mlp.down_proj	16,777,216	583,555,072
model.layers.5.mlp.gate_proj	16,777,216	600,332,288
model.layers.5.mlp.up_proj	16,777,216	617,109,504
model.layers.5.post_attention_layernorm	2,048	617,111,552
model.layers.5.self_attn.k_proj	1,048,576	618,160,128
model.layers.5.self_attn.o_proj	4,194,304	622,354,432
model.layers.5.self_attn.q_proj	4,194,304	626,548,736
model.layers.5.self_attn.v_proj	1,048,576	627,597,312
model.layers.6.input_layernorm	2,048	627,599,360
model.layers.6.mlp.down_proj	16,777,216	644,376,576
model.layers.6.mlp.gate_proj	16,777,216	661,153,792
model.layers.6.mlp.up_proj	16,777,216	677,931,008
model.layers.6.post_attention_layernorm	2,048	677,933,056
model.layers.6.self_attn.k_proj	1,048,576	678,981,632
model.layers.6.self_attn.o_proj	4,194,304	683,175,936
model.layers.6.self_attn.q_proj	4,194,304	687,370,240
model.layers.6.self_attn.v_proj	1,048,576	688,418,816
model.layers.7.input_layernorm	2,048	688,420,864
model.layers.7.mlp.down_proj	16,777,216	705,198,080
model.layers.7.mlp.gate_proj	16,777,216	721,975,296
model.layers.7.mlp.up_proj	16,777,216	738,752,512
model.layers.7.post_attention_layernorm	2,048	738,754,560
model.layers.7.self_attn.k_proj	1,048,576	739,803,136
model.layers.7.self_attn.o_proj	4,194,304	743,997,440
model.layers.7.self_attn.q_proj	4,194,304	748,191,744
model.layers.7.self_attn.v_proj	1,048,576	749,240,320
model.layers.8.input_layernorm	2,048	749,242,368
model.layers.8.mlp.down_proj	16,777,216	766,019,584
model.layers.8.mlp.gate_proj	16,777,216	782,796,800
model.layers.8.mlp.up_proj	16,777,216	799,574,016
model.layers.8.post_attention_layernorm	2,048	799,576,064
model.layers.8.self_attn.k_proj	1,048,576	800,624,640
model.layers.8.self_attn.o_proj	4,194,304	804,818,944
model.layers.8.self_attn.q_proj	4,194,304	809,013,248
model.layers.8.self_attn.v_proj	1,048,576	810,061,824
model.layers.9.input_layernorm	2,048	810,063,872
model.layers.9.mlp.down_proj	16,777,216	826,841,088
model.layers.9.mlp.gate_proj	16,777,216	843,618,304
model.layers.9.mlp.up_proj	16,777,216	860,395,520
model.layers.9.post_attention_layernorm	2,048	860,397,568
model.layers.9.self_attn.k_proj	1,048,576	861,446,144
model.layers.9.self_attn.o_proj	4,194,304	865,640,448
model.layers.9.self_attn.q_proj	4,194,304	869,834,752
model.layers.9.self_attn.v_proj	1,048,576	870,883,328
model.layers.10.input_layernorm	2,048	870,885,376
model.layers.10.mlp.down_proj	16,777,216	887,662,592
model.layers.10.mlp.gate_proj	16,777,216	904,439,808
model.layers.10.mlp.up_proj	16,777,216	921,217,024
model.layers.10.post_attention_layernorm	2,048	921,219,072

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Module	Params	Running total
model.layers.10.self_attn.k_proj	1,048,576	922,267,648
model.layers.10.self_attn.o_proj	4,194,304	926,461,952
model.layers.10.self_attn.q_proj	4,194,304	930,656,256
model.layers.10.self_attn.v_proj	1,048,576	931,704,832
model.layers.11.input_layernorm	2,048	931,706,880
model.layers.11.mlp.down_proj	16,777,216	948,484,096
model.layers.11.mlp.gate_proj	16,777,216	965,261,312
model.layers.11.mlp.up_proj	16,777,216	982,038,528
model.layers.11.post_attention_layernorm	2,048	982,040,576
model.layers.11.self_attn.k_proj	1,048,576	983,089,152
model.layers.11.self_attn.o_proj	4,194,304	987,283,456
model.layers.11.self_attn.q_proj	4,194,304	991,477,760
model.layers.11.self_attn.v_proj	1,048,576	992,526,336
model.layers.12.input_layernorm	2,048	992,528,384
model.layers.12.mlp.down_proj	16,777,216	1,009,305,600
model.layers.12.mlp.gate_proj	16,777,216	1,026,082,816
model.layers.12.mlp.up_proj	16,777,216	1,042,860,032
model.layers.12.post_attention_layernorm	2,048	1,042,862,080
model.layers.12.self_attn.k_proj	1,048,576	1,043,910,656
model.layers.12.self_attn.o_proj	4,194,304	1,048,104,960
model.layers.12.self_attn.q_proj	4,194,304	1,052,299,264
model.layers.12.self_attn.v_proj	1,048,576	1,053,347,840
model.layers.13.input_layernorm	2,048	1,053,349,888
model.layers.13.mlp.down_proj	16,777,216	1,070,127,104
model.layers.13.mlp.gate_proj	16,777,216	1,086,904,320
model.layers.13.mlp.up_proj	16,777,216	1,103,681,536
model.layers.13.post_attention_layernorm	2,048	1,103,683,584
model.layers.13.self_attn.k_proj	1,048,576	1,104,732,160
model.layers.13.self_attn.o_proj	4,194,304	1,108,926,464
model.layers.13.self_attn.q_proj	4,194,304	1,113,120,768
model.layers.13.self_attn.v_proj	1,048,576	1,114,169,344
model.layers.14.input_layernorm	2,048	1,114,171,392
model.layers.14.mlp.down_proj	16,777,216	1,130,948,608
model.layers.14.mlp.gate_proj	16,777,216	1,147,725,824
model.layers.14.mlp.up_proj	16,777,216	1,164,503,040
model.layers.14.post_attention_layernorm	2,048	1,164,505,088
model.layers.14.self_attn.k_proj	1,048,576	1,165,553,664
model.layers.14.self_attn.o_proj	4,194,304	1,169,747,968
model.layers.14.self_attn.q_proj	4,194,304	1,173,942,272
model.layers.14.self_attn.v_proj	1,048,576	1,174,990,848
model.layers.15.input_layernorm	2,048	1,174,992,896
model.layers.15.mlp.down_proj	16,777,216	1,191,770,112
model.layers.15.mlp.gate_proj	16,777,216	1,208,547,328
model.layers.15.mlp.up_proj	16,777,216	1,225,324,544
model.layers.15.post_attention_layernorm	2,048	1,225,326,592
model.layers.15.self_attn.k_proj	1,048,576	1,226,375,168
model.layers.15.self_attn.o_proj	4,194,304	1,230,569,472
model.layers.15.self_attn.q_proj	4,194,304	1,234,763,776
model.layers.15.self_attn.v_proj	1,048,576	1,235,812,352
model.norm	2,048	1,235,814,400